Bottom-Up Return Forecasting under an Explicit Alpha–Risk–Cost Objective

A historical quantitative equity methodology study informed by Acadian Asset Management's 2026 Investor Forum

Research period: February 2020–December 2025

1. Introduction and Research Motivation

A systematic investment process must reconcile three difficult and competing goals: extracting security-level expected returns from imperfect information, preventing incidental common-factor exposures from overwhelming selection risk, and preserving forecast return when translating hypothetical positions into traded holdings. Acadian Asset Management's public May 2026 Investor Forum describes this integrated investment architecture and explicitly differentiates known market risks, generic style premia, intentional alpha risks, and implementation risks. This report investigates an independent implementation of those principles using only openly available historical observations.

The core research question is whether the explicit inclusion of systematic factor risk and modeled liquidity/impact costs materially changes the allocations, turnover, realized net results, and inferential conclusions of a bottom-up stock-return forecasting process. The project is designed so that a negative empirical result remains a valid research outcome: forecasts are not presumed to work, risk management is not presumed to increase realized returns, and transaction-cost penalties are not presumed to create alpha.

2. Research Objective, Hypotheses, and Scope

The primary objective is to implement and assess an interpretable forecast–risk–cost decision system. Secondary objectives are to verify temporal data integrity, estimate risk decomposition under observed market factors, conduct matched benchmark comparisons, quantify modeled trading frictions, test statistical significance, and create a reproducible Python codebase suitable for technical scrutiny.

Hypothesis	Test or observable implication
H1: Forecast skill	Mean monthly cross-sectional Spearman information coefficient should be positive.
H2: Factor-aware controls	Active factor exposure and realized drawdown/volatility should improve relative to alpha-only allocations.
H3: Transaction-cost objective	Turnover and estimated implementation costs should fall after penalizing trading.
H4: Active performance	Net performance relative to the matched basket benchmark should be reliably positive under HAC inference.

Scope is deliberately restricted to a research sample of liquid, currently surviving S&P 500 constituents. This is not the global equity universe mentioned by Acadian, and without stock-level financial statements the project cannot reconstruct value, earnings, quality, growth, or country–industry forecasts. It instead tests technical, statistical peer, market-regime and liquidity-derived signals and identifies exactly where a fundamentals extension would connect to the architecture.

3. Relation to Acadian's Published Investment Process

The source presentation identifies three forecasting layers—security-specific, overlooked peer groups, and macro group forecasts—and integrates them with dynamic cost models and risk-aware construction. Its risks taxonomy distinguishes broad market beta, FX, rates and small-cap exposures from generic style premia and proprietary alpha risks. Because neither proprietary signals nor internal risk files are public, this report references the documented architecture while implementing different, reproducible research estimators.

Public disclosure component	This independent implementation	Missing from supplied evidence

Stock-specific thematic forecasts	Price momentum, reversal, range, volatility, liquidity	Value, quality, earnings, analyst revisions
Peer networks	Top-five trailing return-correlation neighbors	Economic supply-chain peer identification
Macro / group forecasts	Past French market trend, industry dispersion interactions	Geography, country–industry intersection
Risk categories	Six-factor beta limits plus specific risk variance	Acadian proprietary alpha-risk model
Dynamic costs / trading	ADV, volatility and range-based nonlinear trading penalty	Actual bid/ask spreads, broker selection and trade fills

Public source: Acadian Asset Management Inc., Investor Forum, 19 May 2026, slides 19–23 (SEC Exhibit 99.2). No proprietary alpha, traded-fund performance or affiliation is claimed.

4. Data Sources, Coverage, and Provenance

Input	Observed sample / features	Role
Aethereal current-member S&P 500	1,444,124 stock-days; 503 tickers; 2015-01-02 to 2026-11-10	Adjusted prices, OHLCV, ADV, vol, open-to-open realized returns
Fama/French 5 factors (2×3)	Mkt–RF, SMB, HML, RMW, CMA, RF; monthly	Market and common-factor risk estimation
Fama/French Momentum	MOM; monthly	Generic momentum exposure monitoring
49 Industry Portfolios	Monthly value-weighted industry returns	Past industry dispersion / macro-state proxy

The French downloads identify the September 2026 CRSP vintage. Factor returns are supplied in percent, converted to decimals before analysis. The parser reads the value-weighted monthly industry section only. Source artifacts and SHA-256 checksums are stored with the repository. The market-factor dataset is not a direct, tradable S&P 500 index benchmark; performance comparisons primarily use a dynamic equal-weight basket of the same 40 selected names.

Crucially, the vendor states that its price panel includes only companies belonging to the S&P 500 at the download date. Historical removals, acquisitions, bankruptcies, and delistings are absent. Thus this experiment is a survivor-universe stock-selection backtest and should not be described as a survivorship-bias-free historical S&P 500 strategy.

5. Experimental Timing, Universe, and Information Set

All signals are computed after the final trading close of month t. The portfolio is modeled as traded at the first opening quote of month t+1, then held until the first opening quote of month t+2. Because only an unadjusted opening quote and adjusted closing quote are available, the adjusted-open approximation multiplies raw open by adj_close/close on the same date. Dividend and split adjustments are vendor-derived and could reflect subsequent data revisions.

$$R_{i,t+1} = \frac{O(i,t+2)}{O(i,t+1)} - 1$$

The security sample is reconstituted monthly using trailing 20-session dollar average daily volume, without using future return data. A stock must have at least 252 previous daily observations and a complete feature vector; the 40 most liquid eligible survivors enter the sample. The matched benchmark is monthly rebalanced equal-weight in these same 40 names, and modeled trading costs are deducted from this benchmark as well. The 71 test decisions are January 2020–November 2025; the associated holding windows cover approximately February 2020–December 2025.

6. Feature Engineering and the Factor Library

Security features transform raw prices and activity into comparable, interpretable signals. Features are calculated from historical dates only and transformed each month into within-date percentile ranks centered on zero. This preserves ordinal information, reduces the impact of outliers, and prevents naturally high-valued features such as dollar volume from dominating the Ridge penalty.

Feature	Construction	Expected information channel
12–2 month momentum	$P_{(t-1)}/P_{(t-12)} - 1$	Persistent relative return trends
6–2 month momentum	$P_{(t-1)}/P_{(t-6)} - 1$	Intermediate trend persistence
One-month reversal	$-monthly\ close/close\ return\ at\ t$	Short-horizon correction
Low volatility	$-std\ (daily\ returns, 63) \times \sqrt{21}$	Defensive equity characteristics
Low intraday range	$-median\ ((high - low)/close, 20)$	Volatility / liquidity-related price behavior
Dollar-volume liquidity	$log\ (mean\ (close \times volume, 20))$	Trading capacity and market interest
Historical peer momentum	Mean momentum of five highly correlated past peers	Cross-security diffusion proxy
Market-regime interaction	$\mathrm{Mom}_6 \times \mathrm{past}_6 - month\ market\ excess - return\ sum$	Conditional trend behavior
Industry-dispersion interaction	$\mathrm{Mom}_6 \times \mathrm{past}_3 - month\ industry\ return\ dispersion$	Cross-industry environment response

The peer model is based on the five highest trailing-12-month return correlations within 100 contemporaneously liquid candidates, requiring at least eight paired historical observations. It does not infer economic supply-chain links or proprietary information networks. Stand-alone market beta is deliberately omitted as an expected-return signal; it is monitored in the risk model instead. Real fundamental value, earnings and growth signals would require financial statement observation dates and as-of availability timestamps, and are not silently synthesized here.

7. Stock-Level Forecasting: Rolling Regularized Regression

Each rebalance trains a pooled cross-sectional/time-series Ridge regression on previously completed signal–outcome observations from approximately the preceding 48 months. Features are standardized using training data only, and an L2 penalty α=100 stabilizes a moderately correlated feature set. Forecasts are bounded to ±15% per monthly holding period. No return from the test holding period enters model fitting, and the procedure is repeated at every decision month.

$$\beta^{Ridge} = arg\ min\ \Sigma\ (\ y_{i,t+1} - \beta_0 - x^T\beta\ )^2\ +\ \lambda\ \|\beta\|^2$$

$$\hat{\alpha}_{i,t+1}\ =\ clip(\ \beta_0 + x^T\beta^{Ridge}, -0.15, 0.15)$$

The prediction target is the next open-to-open monthly holding return, rather than a return ending at the signal close. Spearman rank information coefficient is recorded each test month for the 40-name cross-section. Regression produces forecasts, not proof of persistent alpha; forecast IC, coefficient stability, and sensitivity of realized outcomes are explicitly reviewed.

8. Factor Risk Model and Covariance Shrinkage

For each selected stock, historical monthly returns are regressed against six published Fama/French risk factors: market excess return, size, value, profitability, investment, and momentum. The 36-month window generates a factor-loading vector for each stock and a residual variance. The latter is floored at 0.0004 in monthly squared-return units. Factor covariances are estimated using the Ledoit–Wolf shrinkage estimator, reducing numerical instability associated with a short monthly sample.

$$r_{i,t} - r_{f,t} = a_i + \beta^T f_t + \varepsilon_{i,t}$$

$$\Sigma = B\,\Sigma_F\,B^T + diag(\sigma^2)$$

The active covariance penalty is computed around the equal-weight basket b: $activerisk^2 = (w-b)^{\mathrm{T}}\Sigma(w-b)$. This is conceptually distinct from total portfolio variance. Explicit linear constraints limit active market loading to ±0.15 and each active generic style-factor loading to ±0.30. Model residual variance is not labeled proprietary alpha risk: omitted systematic exposures may reside there, and a covariance model does not create positive return forecasts.

9. Transaction Costs and Capacity-Dependent Market Impact

The available price file lacks quoted bid–ask spreads, realized execution prices, commissions, order book imbalance, or measured market impact. Therefore a transparent two-component cost scenario is constructed from observed daily price ranges, volatility and ADV. The modeled half-spread-like term uses one-quarter of a trailing median daily high–low range, explicitly not an empirical quoted spread. The impact term grows superlinearly with trade weight (power 3/2) as market participation increases.

$$TC(\Delta w) = \Sigma\,[\,0.25 \times Range_{20} \times abs(\Delta w) + 0.08 \times \sigma_m \times \sqrt{(AUM\,/\,ADV_{20})} \times abs(\Delta w)^{3/2}\,]$$

The base scenario assumes USD 10 million invested capital. An additional 50 basis-point charge on the weight of names leaving the universe approximates liquidation costs; actual costs might be materially higher or lower. The function is differentiable away from zero weight changes and its analytical gradient is checked against finite differences in the software tests. Fixed-holdings cost multipliers of 0, 0.5, 1, 1.5, 2 and 3 illustrate implementation sensitivity but are not optimization stress tests.

10. Constrained Alpha–Risk–Cost Optimization

Let w be target stock weights, b equal weights in the current basket, w_0 drifted pre-trade weights approximated from known month-end prices, $\hat{\alpha}$ forecast monthly returns and $\hat{\Sigma}$ forecast monthly covariance. The cost-aware portfolio maximizes expected selected-stock return net of risk and the explicit cost function. A continuous SLSQP optimizer uses analytic gradients and portfolio constraints. The risk-aversion setting is λ=12, with no claim of a universal economically optimal calibration.

$$max_w\ \hat{\alpha}^{\mathrm{T}}\,w - \frac{\lambda}{2}\,(w-b)^{\mathrm{T}}\,\Sigma\,(w-b) - TC(w-w_0)$$

$$\Sigma\,w_i = 1; \quad 0 \le w_i \le 0.10;\ \ abs[(w-b)^{\mathrm{T}}\,B_k] \le c_k$$

The benchmark holds equal weights. Alpha-only maximizes forecasts subject to long-only 10% weight limits and a small quadratic stabilization term; it omits explicit factor caps. Alpha+risk incorporates active covariance and exposure caps but ignores trading costs during optimization. Cost-aware uses both; all four portfolios nevertheless pay ex-post estimated trading costs. This isolates a methodological comparison but not a causal estimate of Acadian's performance.

11. Backtest Engine and Portfolio Accounting

At each decision month the procedure selects the liquid basket, fits and forecasts next-window returns, estimates factor exposures and risk, optimizes three alternatives, records desired holdings, computes expected costs, and then credits actual subsequent open-to-open returns from the dataset. Prior holdings are marked using the next month's observed open-to-close return before the next month-end rebalance; future next-opening execution returns are never used to choose next holdings. Positions are subsequently liquidated/rebalanced as necessary. This uses month-end observable drift as an approximation for the following first-opening pre-trade holdings, so unmodeled overnight gaps remain a limitation. If a required next holding return is missing, the engine raises an error rather than treating it as zero or silently reweighting survivors.

$$R_{net,t+1} = (1 - TC_t)(1 + w^T R_{t+1}) - 1$$

Outputs include every monthly strategy result, 2,840 security forecasts, 11,360 stock/strategy holdings rows, factor exposures, training-window audit files, JSON summary, statistical-inference JSON, and ten publication-quality figures. Sharpe ratios presented in the code summary use a zero-risk-free denominator convention and are explicitly labeled; factor-regression studies subtract the aligned French one-month risk-free return.

12. Out-of-Sample Portfolio Performance

Table 5 reports the complete 71-period result after modeled entry and exit cost deductions. The matchable benchmark is a dynamically rebalanced, 40-name equal-weight basket. Since the test consists of present-day index survivors, these CAGRs must not be interpreted as what an investor in January 2020 could necessarily have achieved.

Strategy	Net CAGR	Ann. volatility	Max drawdown	Avg. monthly turnover
Equal-weight basket	15.51%	21.81%	-32.99%	20.08%
Alpha only	7.67%	27.45%	-44.55%	55.01%
Alpha + risk	9.10%	23.07%	-41.25%	52.06%
Alpha + risk + costs	14.97%	23.88%	-41.80%	19.80%

The cost-aware variant records 14.97% CAGR against 15.51% for its benchmark, and a worse maximum drawdown (−41.80% versus −32.99%). By contrast, its average one-way monthly turnover of 19.80% is substantially lower than 55.01% for the alpha-only optimizer and 52.06% for the risk-aware optimizer. These results show material implementation effects; they do not establish superior realized risk control or return generation.

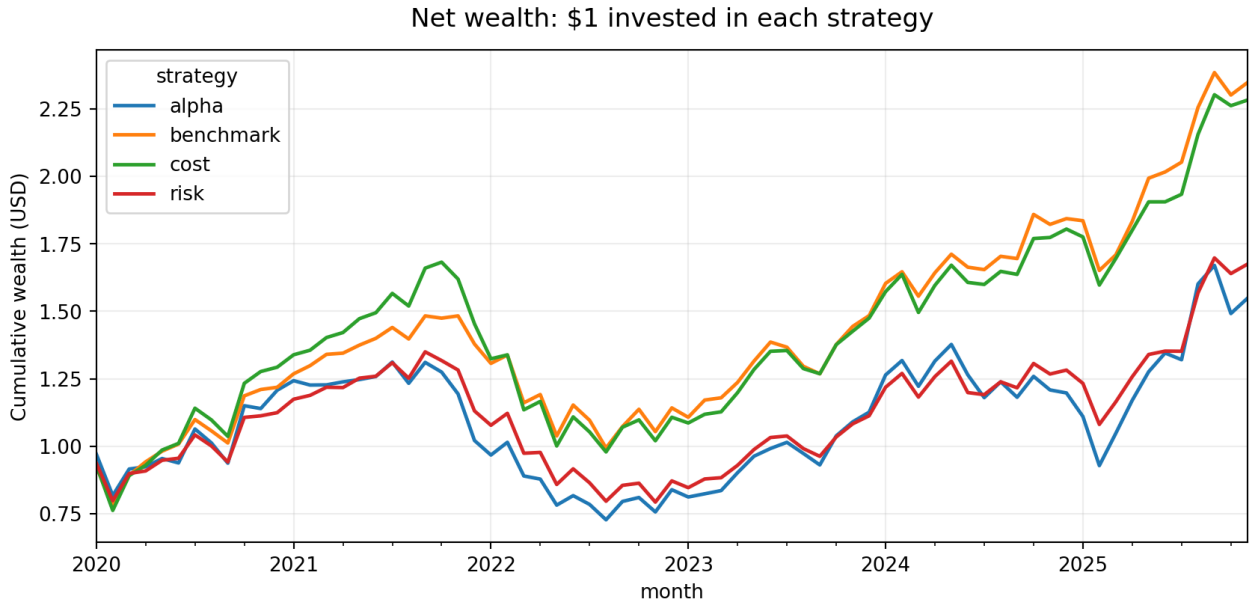


Figure 1. Cumulative net wealth. Monthly modeled net returns; signal month is plotted, holding occurs next month.

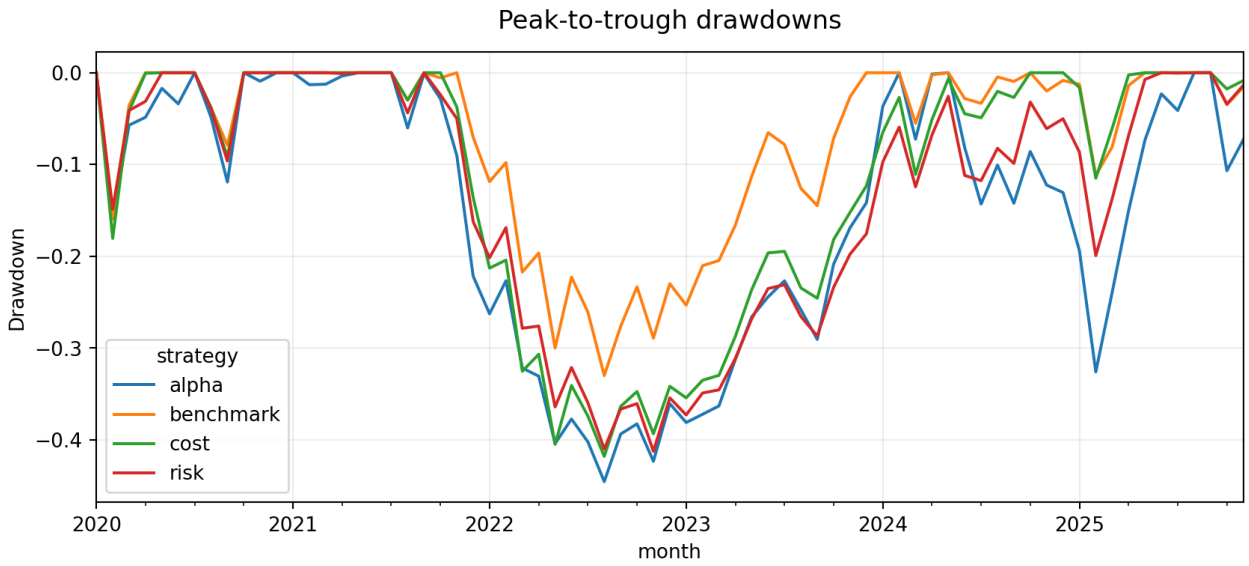


Figure 2. Drawdowns. Peak-to-trough declines computed from the cumulative net wealth series.

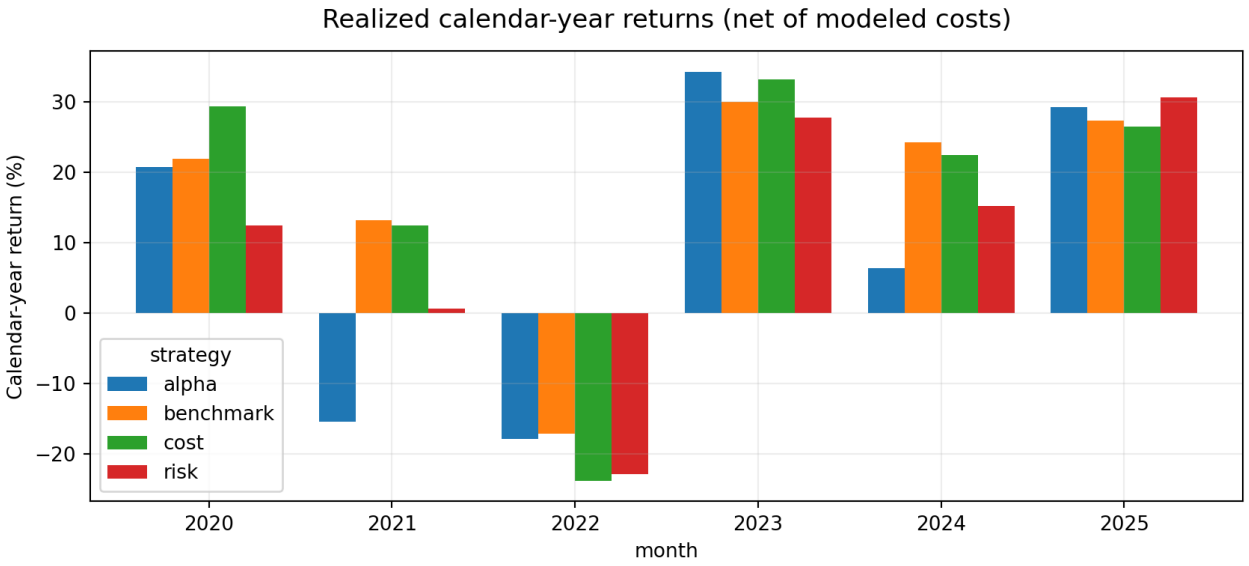


Figure 3. Annual performance. Grouped by decision year; calendar holding periods are shifted approximately one month.

13. Forecast Diagnostics and Statistical Inference

The recorded mean out-of-sample Spearman stock-rank information coefficient equals -0.0193; HAC/Newey–West t=-0.82, p=0.410. A positive rank IC occurs in 35 of 71 months. This fails to support the proposed positive-return-forecast hypothesis. The correct inference is not merely that historical return trends can be modeled, but that the implemented set of public technical and peer proxies has not demonstrated reliable stock ranking skill in this sample.

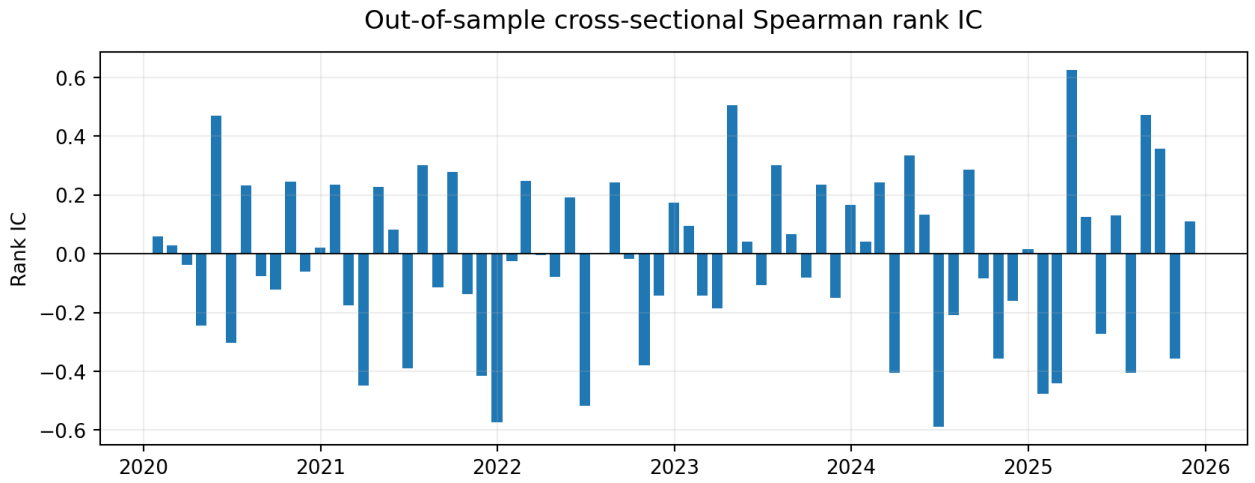


Figure 4. Monthly cross-sectional IC. Spearman correlation of 40 stock forecasts with subsequent realized open-to-open returns.

Portfolio	Net arithmetic active mean / year	Information ratio	HAC p-value
alpha	-5.79%	-0.440	0.251
risk	-5.49%	-0.795	0.027
cost	-0.00%	-0.000	0.999

For the cost-aware portfolio, the mean arithmetic active return is -0.00% annualized, with p=0.999. The HAC 95% confidence interval for the monthly mean active return is [-0.394%, 0.393%]. These estimates make a positive economic alpha conclusion unsupportable. The historical monthly difference is further measured by factor regressions aligned with the following holding month, not mechanically compared with factors from the signal month.

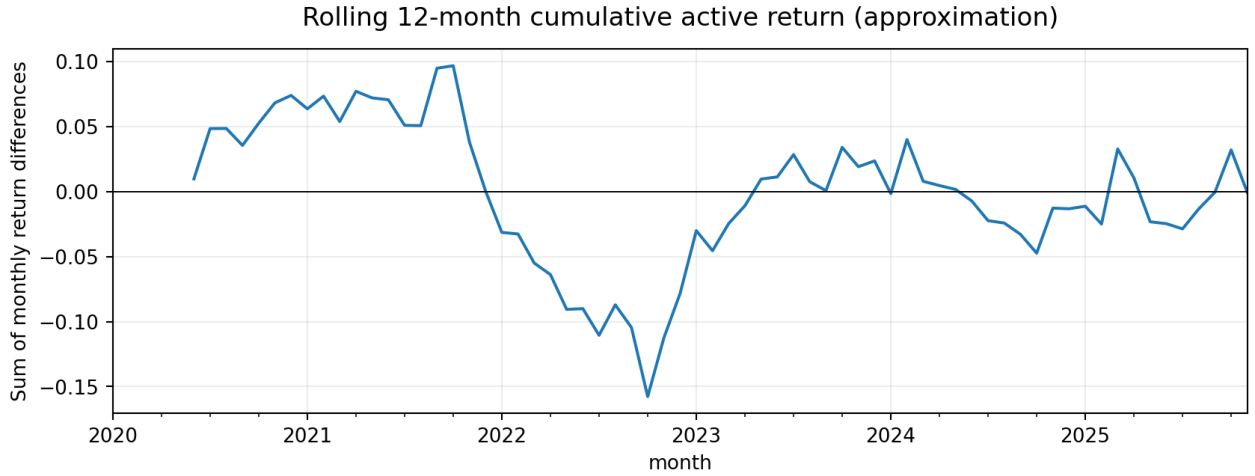


Figure 5. Rolling active return. Sum of next-period active monthly return differences over trailing 12 decision records.

14. Factor-Model Return Attribution and Exposure Control

Portfolio returns net of the aligned next-period risk-free rate are regressed on six monthly Fama/French factors. The estimated intercept is multiplied by 12 for approximate annualized arithmetic presentation. The regression uses HAC standard errors to recognize serial dependence and heteroskedasticity. A regression intercept is a descriptive residual, not evidence of proprietary alpha or omitted-asset returns.

Strategy	Annualized six-factor intercept	Intercept HAC p-value	Model R^2
benchmark	-0.14%	0.945	0.910
alpha	-4.83%	0.380	0.659
risk	-4.70%	0.177	0.811

cost	-1.34%	0.695	0.849

The cost-aware portfolio's six-factor intercept is not statistically significant. Its factor exposures are monitored as signed active betas, centered on the current equal-weight benchmark. The risk model therefore treats market and style premia as exposures to control while leaving expected selection return to the alpha model. The framework still cannot separate proprietary alpha from statistically specific residual risk.

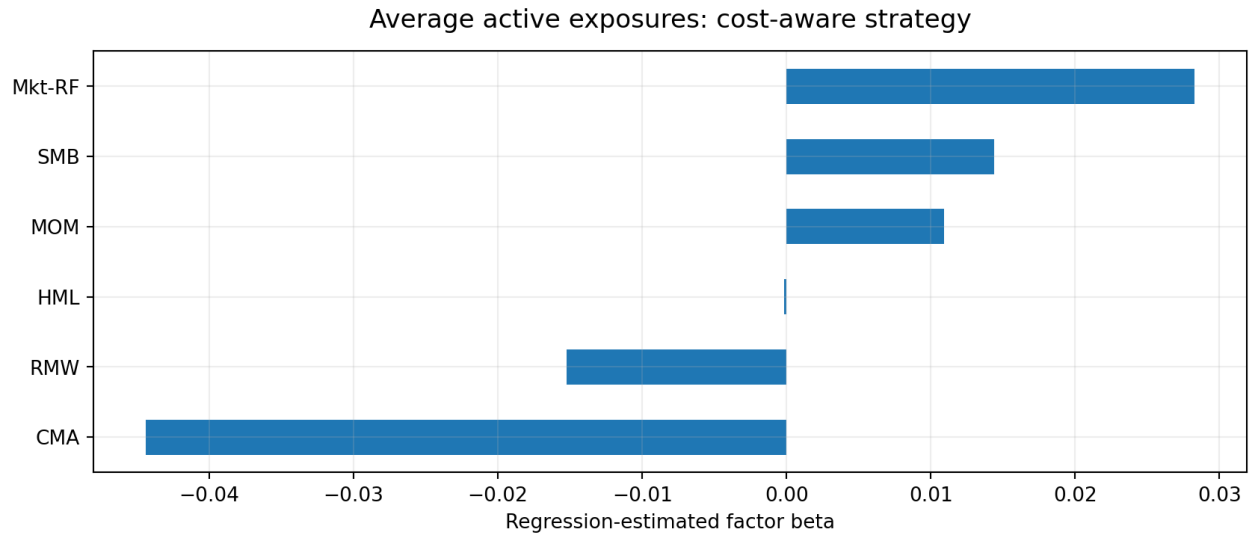


Figure 6. Mean active factor exposures. Mean estimated factor betas over all decisions for the cost-aware portfolio.

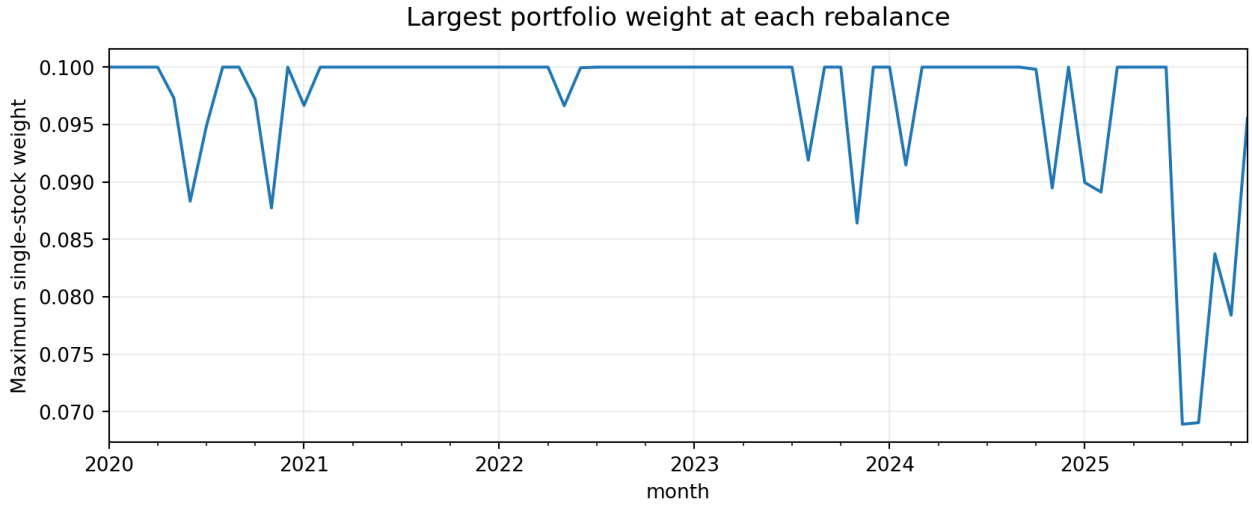


Figure 7. Largest single-name weight. Constrained allocation respects the 10% maximum weight.

15. Trading Frictions, Turnover, and Stress Tests

Transaction costs alter the optimizer's trade-off between crossing a forecast ranking and preserving capital through patient portfolio adjustment. Since prices are daily OHLCV rather than transaction-level bid/ask data, the cost model is deliberately conservative and interpretable rather than empirically calibrated. Frictions are paid by all strategies; the benchmark is not credited with frictionless trading.

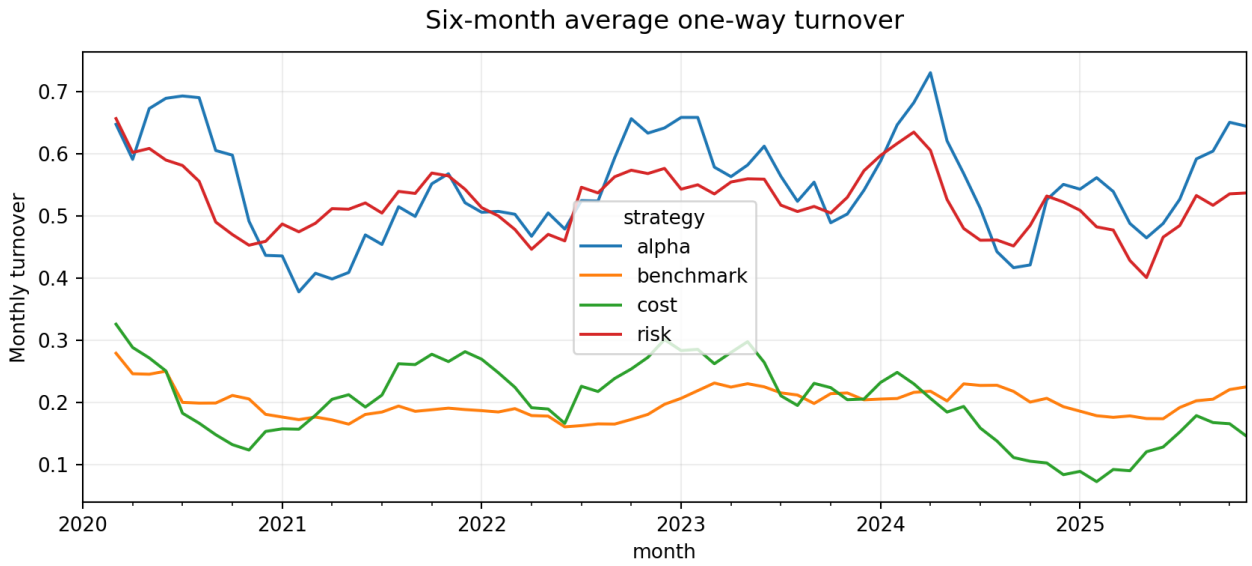


Figure 8. Six-month average turnover. One-way turnover includes orders required to liquidate removed holdings.

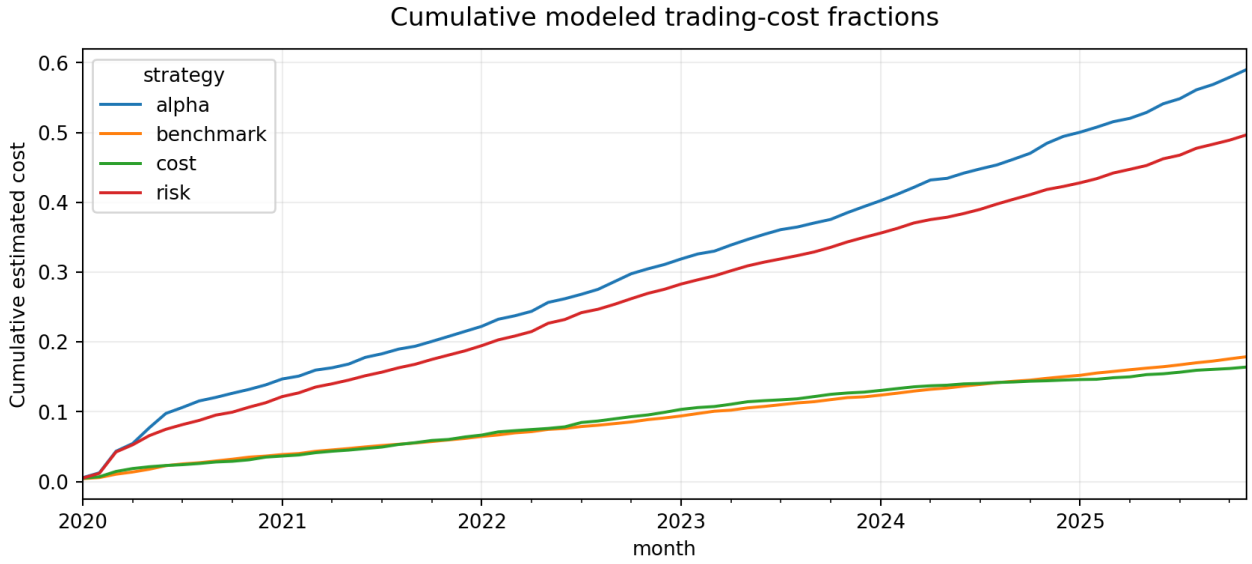


Figure 9. Cumulative model-estimated trading costs. Sum of estimated monthly transaction cost fractions; not a broker fill record.

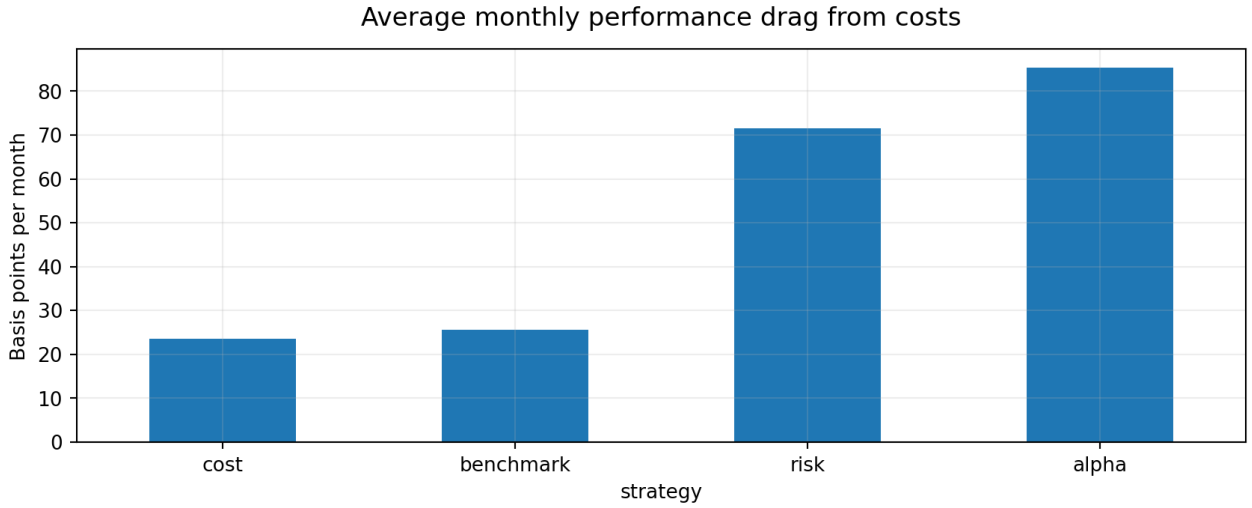


Figure 10. Realized return impact of modeled costs. Mean gross-minus-net monthly performance gap, displayed in basis points.

Multiplier × assumed costs	Benchmark CAGR	Cost-aware CAGR
0×	19.06%	18.21%

0.5×	17.28%	16.58%
1×	15.51%	14.97%
1.5×	13.78%	13.38%
2×	12.06%	11.81%
3×	8.70%	8.73%

Sensitivity holds the base solution weights fixed and changes only post-trade accounting. It cannot answer how an optimizer would change target holdings if fees, the estimated impact coefficient, capital size, or intraday liquidity conditions changed. Full reoptimization, volume participation limits, and executable auction-price simulations would be needed to evaluate capacity at institutional scale.

16. Regime-Level Robustness and Alternative Outcomes

Decision-period split	Equal-weight CAGR	Alpha-only CAGR	Cost-aware CAGR
2020–2022	4.53%	-5.68%	3.46%
2023–2025	28.01%	23.37%	28.14%

The 2020–2022 and 2023–2025 splits demonstrate regime sensitivity but do not constitute independent pre-registered out-of-sample discoveries. The sample spans unusually different volatility and market leadership conditions. The interpretation should account for sample concentration, security survivorship and the choice of liquidity-ranked names, rather than assuming generalization to less liquid, global, or historically delisted assets.

17. Software Architecture, Numerical Verification, and Test Coverage

Python module	Analytical responsibility
data.py	Load and check 4 raw datasets; OHLCV transforms; peer network; historical timestamp labels
models.py	Pooled standardized Ridge forecasts; six-factor OLS; shrinkage covariance
portfolio.py	SLSQP solution with analytic gradients; convex cost scenario; drifted holdings and performance
study.py	Cross-sectional monthly forecast/optimize/execute/report pipeline
reporting.py	HAC inference, factor regression, risk/turnover diagnostics and ten figures
parquet_native.py	Narrow Linux-only offline reader for this provided Parquet source
tests/	Historical timing, constraints, gradients, covariance, files and output invariants

The validation run passed 23 tests. Checks include numeric finite-difference verification of cost gradients; fully invested, long-only and capped holdings; factor exposure caps; positive-definite covariance; French data units; raw data record counts and timestamp order; leakage-sensitive target labels; missing-data refusal; and cross-checks against generated position records. Passing tests does not certify an investment strategy, exchange membership, regulatory compliance, or profitability.

Reproduction instructions: create a Python environment, install the package using `python -m pip install -e .`, rerun `python -m acadian_real.study --root . --end 2025-11`, regenerate inference and figures using `python -m acadian_real.reporting .`, and run `python -m pytest -q`. Source datasets and output files are bundled in the GitHub-ready archive, together with SHA-256 provenance checksums.

18. Limitations and Research Extensions

The dominant weakness is survivorship: source prices include only stocks that still belong to the index in late 2026, so benchmark and signal effects need not survive a point-in-time universe containing historical removals. The absence of point-in-time filing fundamentals prevents value, quality, profitability and earnings forecast reconstruction. French factor histories are the September 2026 *revised* vintage rather than rolling historical

releases. In addition, actual quote spreads, fees, trade execution prices, market-cap data, and broker-level analytics are missing, making the nonlinear cost model a calibrated scenario rather than a verified market-impact estimator.

The next empirical extension requires the point-in-time constituent universe (including delisted securities and their returns), historical security identifiers and sectors, point-in-time SEC/Compustat-like data with release timestamps, and daily quoted effective spread or order-level execution records. With those inputs one can compare classical linear and nonlinear predictors, decomposed value/quality/growth signals, Bayesian risk and covariance estimators, trade-level Almgren–Chriss scheduling, true industry peer networks, capacity tests, purged temporal cross-validation, and nested hyperparameter validation. Those extensions should be implemented only after inputs become available, not represented as current capabilities.

19. Conclusion

The study demonstrates an end-to-end, modular, real-price implementation of a return–risk–cost portfolio construction architecture referenced to a public Acadian disclosure. It connects rolling bottom-up forecasts to a six-factor covariance model, controls market and style exposures explicitly, and models implementation costs in a numerical portfolio optimizer. The outcome does not validate an alpha source: the mean Spearman IC is slightly negative and insignificant, and the fully cost-aware strategy does not outperform the dynamic equal-weight basket on net CAGR or drawdown. Its clearest demonstrated behavior is substantially reduced turnover relative to alpha-only and risk-only optimizers under the chosen model and simulated friction assumptions.

This is a quantitatively meaningful and reproducible methodological contribution, rather than a claim to have reverse engineered Acadian's proprietary process. The restrained conclusion is that a seemingly complete portfolio optimizer cannot compensate for uncertain expected returns, source-universe bias, or imperfect estimates of tradability. The repository provides a transparent baseline for a future genuine point-in-time, fundamentals-rich, and execution-calibrated empirical research paper.

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